

S1. You have been assigned to play the role of an **analyst** and of a **tester** for a new online banking Informatic System (IS). The IS is accessible only by Internet banking from the mobile telephone or via a network of ATM machines (bancomate). The bank clients may ask for information about the bank products both online or directly via call-center operators. **Decide** and **justify** for each item in the list below if this item plays the role of a **stakeholder**, of a **primary actor**, of a **support actor**, if **it doesn't play any role** or if it plays **multiple roles**:

- | | |
|----------------------------------|----------------------------|
| a) The Mobile phone | 0.5 p (0.25 + 0.25) |
| b) The Information System Tester | 0.5 p (0.25 + 0.25) |
| c) The Bank (as Owner of the IS) | 0.5 p (0.25 + 0.25) |
| d) The Bank owner | 0.5 p (0.25 + 0.25) |
| e) The call-center operator | 0.5 p (0.25 + 0.25) |
| f) The bank client (customer) | 0.5 p (0.25 + 0.25) |

Note: the 0.5 p for each item are further divided into 0.25p for the **right decision and 0.25 p for the **right justification**.**

Ex. IS Testers play the role of (0.25p if correct) because(0.25p if justification is correct)

Total for the S1 subject: 3p

S2. Based on the NextGen POS partial model (see the Domain Model slide #4), **propose** a design model based on GRASP design pattern for the following UC realization:

“Customer asks Cashier to remove the item(s) with the lowest price from Sale. System recalculates and displays the new Total Sale amount”.

Note: if there are more than one item with the same low(minimum) price, all these items should be removed from Sale (transaction).

Justify all used design patterns. **Analyze** and **justify** your proposed design with respect to **Low Coupling and High Collision** .

Total for the S2 subject: 3p (1p for the design model, 1p for justifying the design patterns, 1p for justifying Low Coupling and High Collision)

S3. Based on the NextGen POS partial model (see the Domain Model slide #4), **propose** a design model based on the Strategy design pattern for the following UC realization:

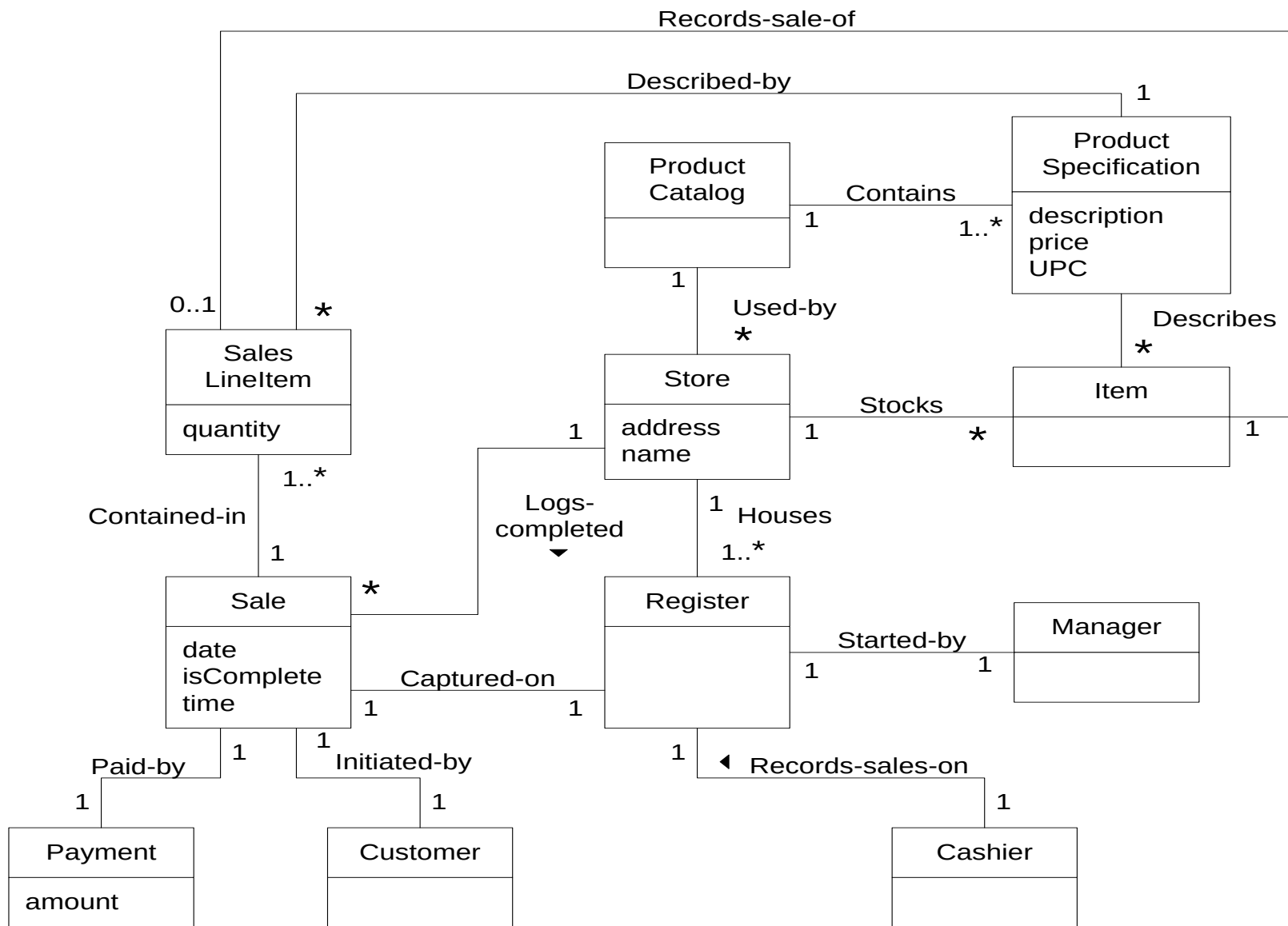
“As part of a promo action, if the total amount of one Sale is more than 200 lei, all supermarket Customers automatically receive at the Cashier’s register a 10% discount from the price of the most valuable purchased item. The System calculates and displays the new total”.

Note: if there are more items with the same maximum price, the discount applies only to one of them.

«**most valuable item**» = item with the highest price

Justify all used design patterns.

Total for the S3 subject: 3p (2p for the design model, 1p for justifying the design patterns)



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